G H PATEL COLLEGE OF ENGINEERING AND TECHNOLOGY

(The Charutar Vidya Mandal (CVM) University)

Department of Information Technology

Program: Computer Science & Engineering (IoT)

VISION

To impart quality education in Information Technology and enable learners to cope with global challenges to build professional career benefitting the sustainable growth of an individual and the society at large.

MISSION

To propose state of the art educational environment equipped with cutting edge technology in the area of Information Technology.

To facilitate learners and faculties with every single opportunity of professional progression embedded in academic scenario itself which can cause enriched workforce contributing to the development of the nation.

To fulfill the noble cause of educating budding technocrats by accelerating the momentum of research and implementing innovative inputs in teaching-learning.

PROGRAMME EDUCATIONAL OBJECTIVES (PEOS)

1. *To build a strong foundation in Computer Science and IoT using modern tools and technologies, while enhancing communication and professional skills.*
2. *To identify real-world problems and develop efficient, cost-effective IoT solutions through systematic analysis, addressing industry and societal needs.*
3. *To grow as ethical professionals with a commitment to teamwork, continuous learning, and adapting to emerging technologies.*

PROGRAMME OUTCOMES (POS)

1. *Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.*
2. *Problem analysis: Identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.*
3. *Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.*
4. *Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.*
5. *Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with*

an understanding of the limitations.

6. *The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.*
7. *Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.*
8. *Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.*
9. *Individual and teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.*
10. *Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.*
11. *Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.*
12. *Life-long learning: Recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.*

PROGRAMME SPECIFIC OUTCOMES (PSOs)

1. *An ability to identify and analyze real-world problems, and to design IoT-based solutions using appropriate hardware, software, and ethical practices.*
2. *An ability to develop intelligent, connected applications by integrating programming skills, embedded systems, sensor networks, and cloud/edge computing.*
3. *An ability to apply data analytics, cybersecurity principles, and emerging IoT technologies to build secure, scalable, and innovative systems for diverse domains.*

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